Course Outline

ULAB UNIVERSITY OF LIBERAL ARTS BANGLADESH

Basic Information

Term	**Fall 2025**
Course Title	**Algorithms**
Course Code	**CSE 2201**
Section	**02**
Credit Hours	**3.0**
Class Schedule	**MW: 03:05 pm – 04:25 pm**
Pre-requisite Course	**CSE1201, CSE 1202, CSE 1203**
Program	**B.Sc in CSE**
Faculty	**Atanu Shuvam Roy**
Contact Email	**atanu.shuvam@ulab.edu.bd**
Contact Number	**+8801937545261**
Office and Location	**PA 406**
Counseling/Office Hour	**Monday: 12 PM – 2 PM**
Google Classroom Code	**https://classroom.google.com/c/ODAzMzk0NzcxOTY2?cjc=gvwthyjv**
Number of Lectures	**24**

1.0 Course Description (from syllabus)/Rational:

This course is an introductory undergraduate course on the design and analysis of algorithms. It introduces several important algorithm design techniques and basic algorithms that are interesting both in theory and practice. The basic algorithm design techniques will include divide-and-conquer, dynamic programming, and greedy techniques for optimization. Techniques for asymptotic analysis of algorithm time bounds by the solution of recurrence equations will also be covered. We will apply these design and analysis techniques to derive algorithms for a variety of tasks such as sorting, searching, graph problems, string matching, computational geometry. The course will also cover the relationship between feasible (polynomial-time) computations and infeasible computations. This will include discussion of the famous P vs. NP question.

1.1 Course Objectives:

1. To provide a thorough understanding of a variety of algorithms with real-life applications and the resource requirements;
2. To introduce several important algorithm design techniques as well as basic algorithms that are interesting both from a theoretical and also practical point of view;
3. To enable students to analyze time and space complexities of algorithms;
4. To emphasize on efficient algorithm designing, solving practical problems through algorithmic techniques and data structures to be used in the implementations of algorithms;
5. To expose the students to a variety of techniques that have practical applications, while conducting detailed analysis of the requirements required by the algorithms.

1.2 Course Outcome/CO: (at the end of the course, students will be able to do:)

CO	Description	Delivery Methods and Activities	Assessment Tools	Domain/ level of learning taxonomy
CO1	Acquire fundamental concepts of algorithms	Lecture	Mid-term exam, Final exam,	Cognitive / L2, Affective / L2
CO2	Explain different paradigms of algorithms	Lecture, Discussion	Mid-term exam, Final exam,	Cognitive / L2, Affective / L2
CO3	Analyze the complexity and performance of various algorithms.	Lecture, Discussion, Group Work	Mid-term exam, Final exam,	Cognitive / L3, Affective / L2

1.3 Teaching and Learning Activities (TLA)

TLA1	Interactive discussion using Online/multimedia or whiteboard.
TLA2	Interactive video and/or scenario based presentation
TLA3	Case Study and group discussion
TLA4	Real-life project in a team to apply knowledge on Algorithms

1.4 Course Delivery Plan (include Lab if any)

Week/Lesson (hour)	Discussion Topic & Book Reference	Student Activities during Online and Onsite and TLA	Assessment and Mapping to CO
Wk 1	**Lesson-1**: Introduction to Algorithms. **Lesson-2**: Introduction to Algorithms: pseudocode, program.	Lesson-1 & 2: Online/Onsite discussion; Interactive content e.g. PPT, etc; **TLA1, TLA2**	**CO1**
Wk 2	**Lesson-3**: Complexity, Asymptotic notation **Lesson-4**: Types of Algorithms, Divide and Conquer: Merge sort.	Lesson-3 & 4: Online/Onsite discussion; Interactive content e.g. PPT, etc; **TLA1, TLA2**	**CO1, CO2,CO3**
Wk 3	**Lesson-5**: Hypothesis to calculate Time Complexity **Lesson-6**: Recurrence Relation: Substitution, Recurrence Relation: Recursion-tree, Master Theorem. Divide and Conquer: Quick sort.	Lesson-5 & 6: Online/Onsite discussion; Interactive content e.g. PPT, etc; **TLA1, TLA2, TLA3**	**CO1, CO2, CO3**
Wk 4 1st CT Week	**Lesson-7:** Discussion on Greedy Algorithms, Fractional Knapsack Problem, Activity-Selection Problem. **Lesson-8:** Discussion on Spanning Tree, MST	Lesson-7 & 8: Online/Onsite discussion; Interactive content e.g. PPT, etc; **TLA1, TLA2, TLA3**	**CO1, CO2, CO3**
Wk 5	**Lesson-9**: Review discussion on 1st Quiz, Prims Algorithm, krushkal Algorithm. **Lesson-10**: Discussion on Data Compression: Huffman Code Problem, Huffman's algorithm;	Lesson-9 & 10: Online/Onsite discussion; Interactive content e.g. PPT, etc; **TLA1, TLA2, TLA3**	**CO1, CO2, CO3**
Wk 6	**Lesson 11:** Review discussion of Wk 4,5	Lesson-11 & 12: Online/Onsite discussion;	**CO1, CO2, CO3**

	Lesson 12: Review discussion for Mid Exam;	Interactive content e.g. PPT, etc; **TLA1, TLA2, TLA3**	
Wk 7	**Midterm Exam Week**		
Wk 8	**Lesson 13:** Discussion on Dynamic Programming, Tabulation, Memoization **Lesson 14:** 0/1 knapsack problem, Longest Common Subsequence Algorithm;	Lesson-13 & 14: Online/Onsite discussion; Interactive content e.g. PPT, etc; **TLA1, TLA2, TLA3**	**CO1, CO2, CO3**
Wk 9	**Lesson 15**: Matrix chain multiplication problem and its solution. Review discussion on Lesson 13 and Lesson 14; **Lesson 16**: Discussion on Graph Algorithms, Dijkstra's Algorthm.	Lesson-15 & 16: Online/Onsite discussion; Interactive content e.g. PPT, etc; **TLA1, TLA2, TLA3**	**CO1, CO2, CO3**
Wk 10 2nd CT Week	**Lesson 17:** Discussion on Single Source Shortest Path - BellmanFord Algorithm; **Lesson 18:** Discussion on Single Source Shortest Path - Directed Acyclic Graph;	Lesson-17 & 18: Online/Onsite discussion; Interactive content e.g. PPT, etc; **TLA1, TLA2, TLA3**	**CO1, CO2, CO3** **Class Test# 2**
Wk 11	**Lesson 19:** Discusssion on All pair shortest path – Floyd warshall algorithms. **Lesson 20:** Discussion on NP-completeness;	Lesson-19 & 20: Online/Onsite discussion; Interactive content e.g. PPT, etc; **TLA1, TLA2, TLA3**	**CO1, CO2, CO3**
Wk 12	**Lesson 21:** Overview		**Assignment**
Wk 13	**Final Exam Week**		

1.5 Text and Reference Materials

Text Book(s): Introduction to Algorithms, 3rd edition. T.H. Cormen, C.E. Leiserson, R.L. Rivest, C. Stein

Reference Material/Book(s):

(1) The Art of Computer Programming (Vol-1): Fundamental Algorithms, 3rd edition. Donald E. Knuth

(2) Schlar.google.com and Google search engine

1.6 Distribution of Marks for Assessment

Assessments	%
Mid Term Examination	25
Final Examination	40
Attendance & class performance	10
Assignment & Quizzes	25
Total	100

1.7. Grading Policy

Policy	Letter Grade	Grade Point	Assessments
95% and above	A+	4.00	Outstanding
85% to 94%	A	4.00	Superlative
80% to 84%	A-	3.80	Excellent
75% to 79%	B+	3.30	Very Good
70% to 74%	B	3.00	Good
65% to 69%	B-	2.80	Average
60% to 64%	C+	2.50	Below Average
55% to 59%	C	2.20	Passing
50% to 54%	D	1.50	Probationary
below 50%	F	0.00	Fail
--	I	0.00	Incomplete
--	W	0.00	Withdrawn
--	AW	0.00	Administrative Withdrawal

.... *Course Teacher* Date: 22/09/25	.... *Head of the Department* Date:

Appendix-1: Program outcomes

POs	Category	Program Outcomes
a	**Engineering Knowledge**	Apply the knowledge of mathematics, science, engineering fundamentals and an engineering specialization to the solution of complex engineering problems.
b	**Problem Analysis**	Identify, formulate, research the literature and analyze complex engineering problems and reach substantiated conclusions using first principles of mathematics, the natural sciences and the engineering sciences.
c	**Design/Development of Solutions**	Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for public health and safety as well as cultural, societal and environmental concerns.
d	**Investigations**	Conduct investigations of complex problems, considering design of experiments, analysis and interpretation of data and synthesis of information to provide valid conclusions.
e	**Modern tool usage**	Create, select and apply appropriate techniques, resources and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
f	**The engineer and society**	Apply reasoning informed by contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to professional engineering practice.
g	**Environment and sustainability**	Understand the impact of professional engineering solutions in societal and environmental contexts and demonstrate the knowledge of, and need for sustainable development.
h	**Ethics**	Apply ethical principles and commit to professional ethics, responsibilities and the norms of the engineering practice.
i	**Individual work and teamwork**	Function effectively as an individual and as a member or leader of diverse teams as well as in multidisciplinary settings.
j	**Communication**	Communicate effectively about complex engineering activities with the engineering community and with society at large. Be able to comprehend and write effective reports, design documentation, make effective presentations and give and receive clear instructions.
k	**Project management and finance**	Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work as a member or a leader of a team to manage projects in multidisciplinary environments.
l	**Life Long Learning**	Recognize the need for and have the preparation and ability to engage in independent, life-long learning in the broadest context of technological change.